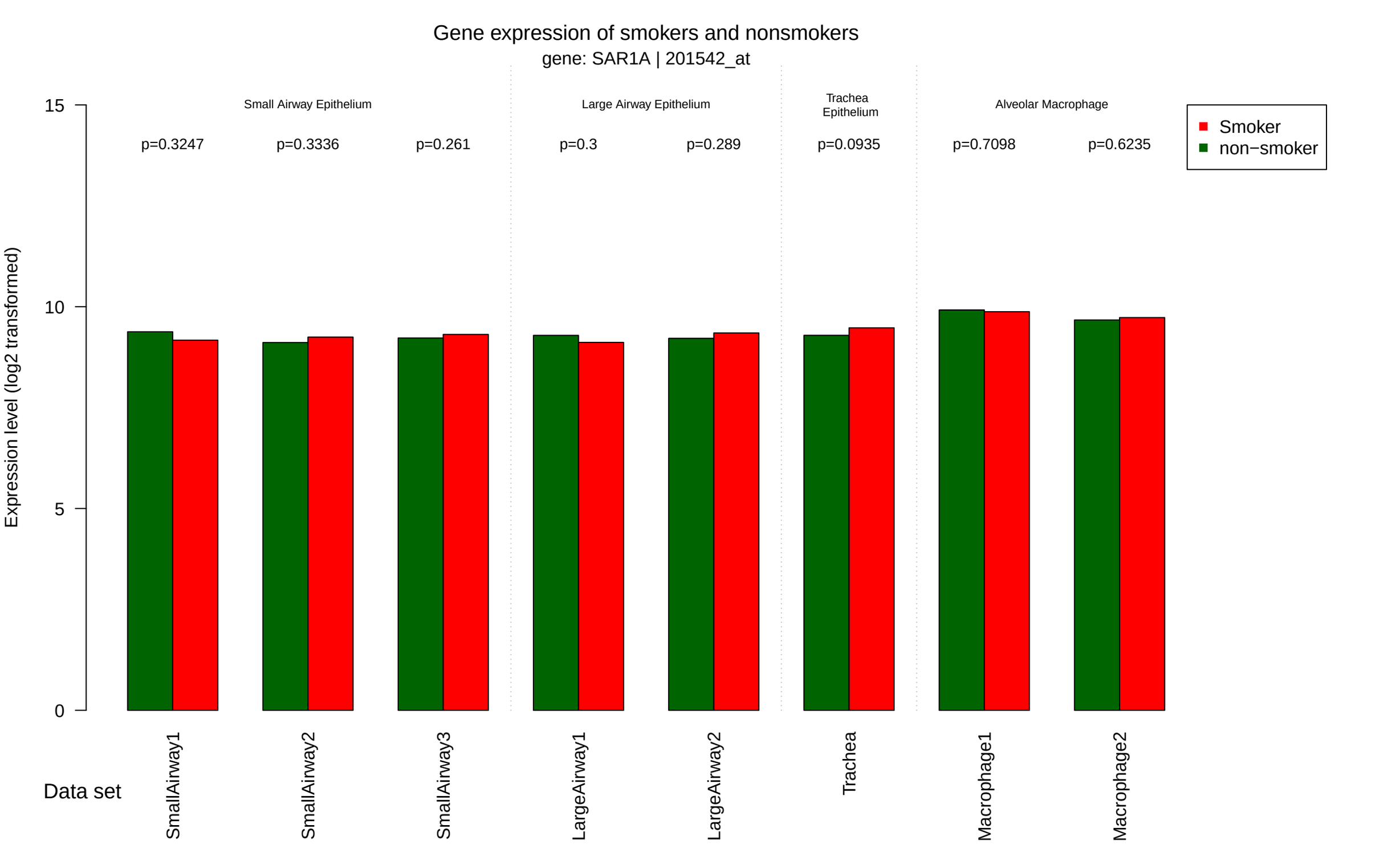
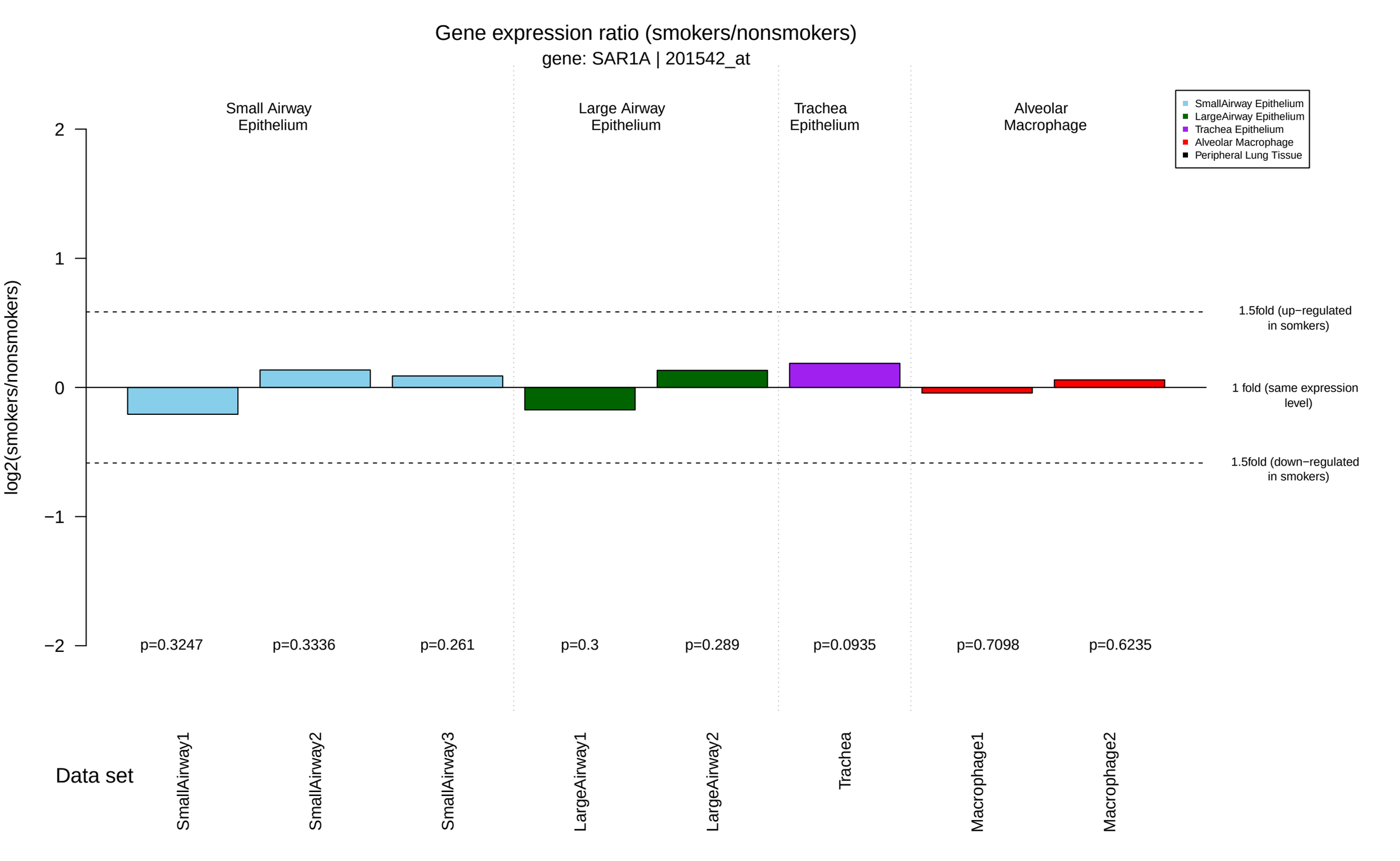
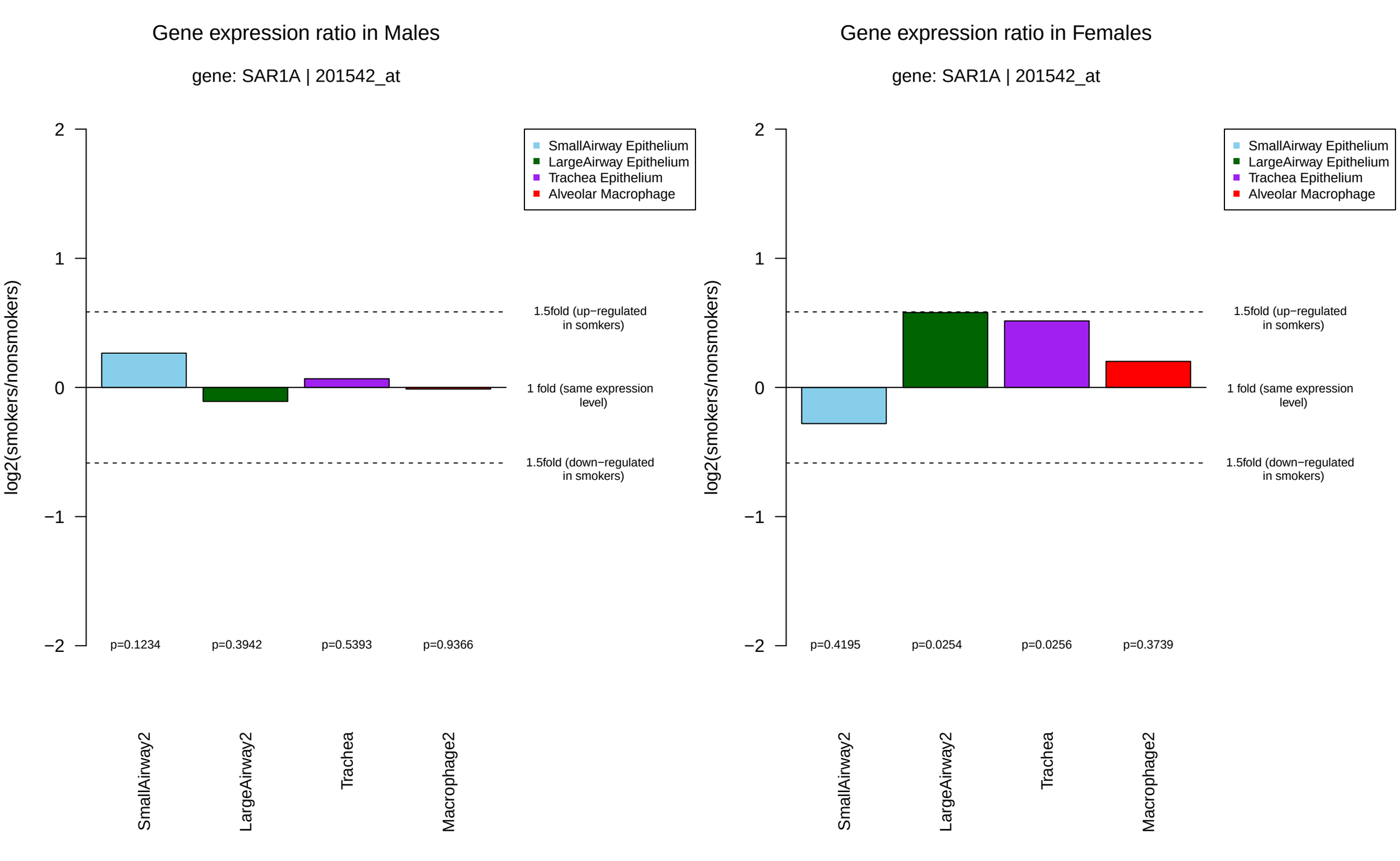
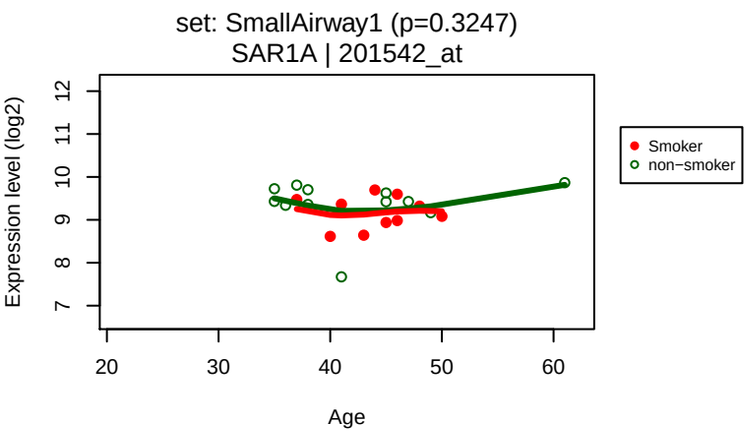
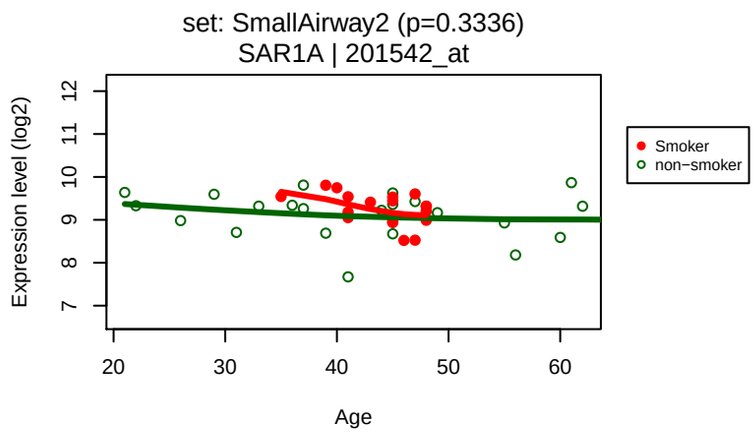
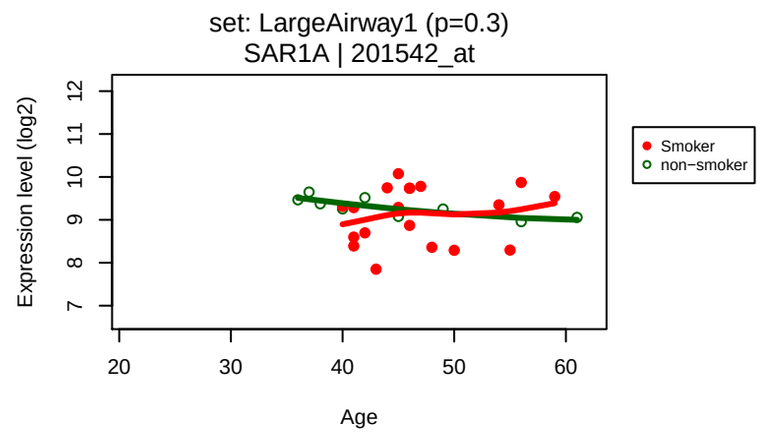

Gene expression of smokers and nonsmokers
gene: SAR1A | 201542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3247
p=0.3336
p=0.261
p=0.3
p=0.289
p=0.0935
p=0.7098
p=0.6235
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

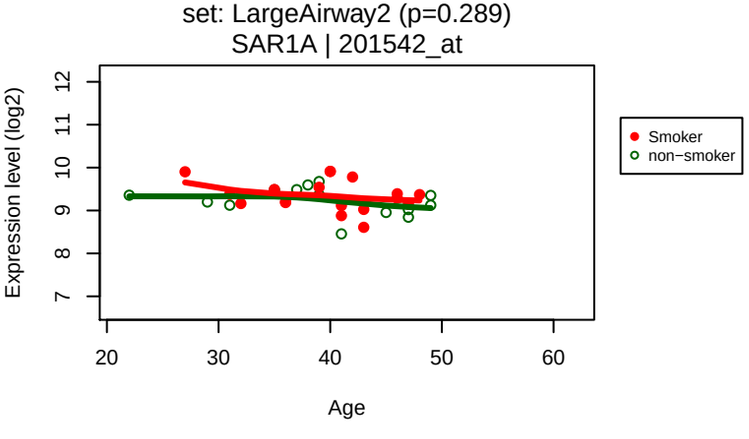

Gene expression ratio (smokers/nonsmokers)
gene: SAR1A | 201542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3247
p=0.3336
p=0.261
p=0.3
p=0.289
p=0.0935
p=0.7098
p=0.6235
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

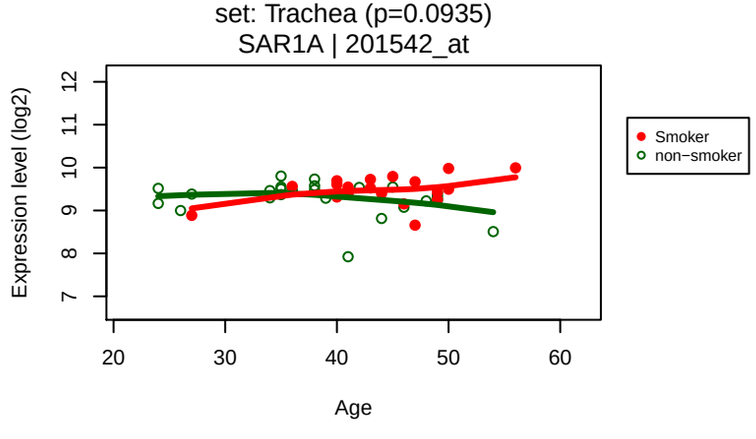

Gene expression ratio in Males
gene: SAR1A | 201542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1234
p=0.3942
p=0.5393
p=0.9366
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAR1A | 201542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4195
p=0.0254
p=0.0256
p=0.3739
SmallAirway2
LargeAirway2
Trachea
Macrophage2

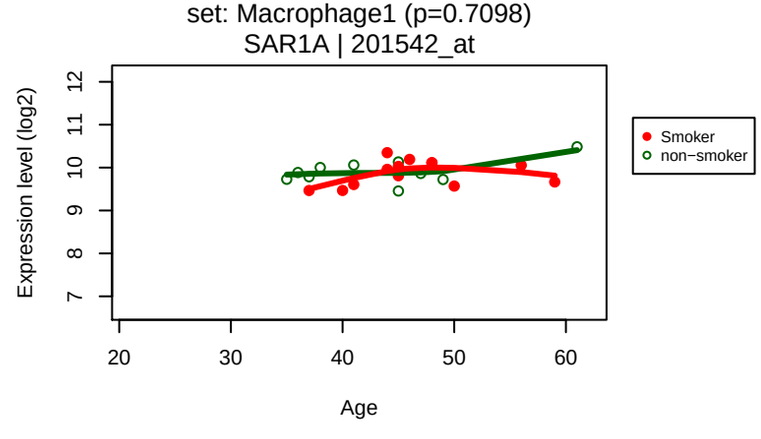

set: SmallAirway1 (p=0.3247)
SAR1A | 201542_at
Expression level (log2)
Age
Smoker
non−smoker

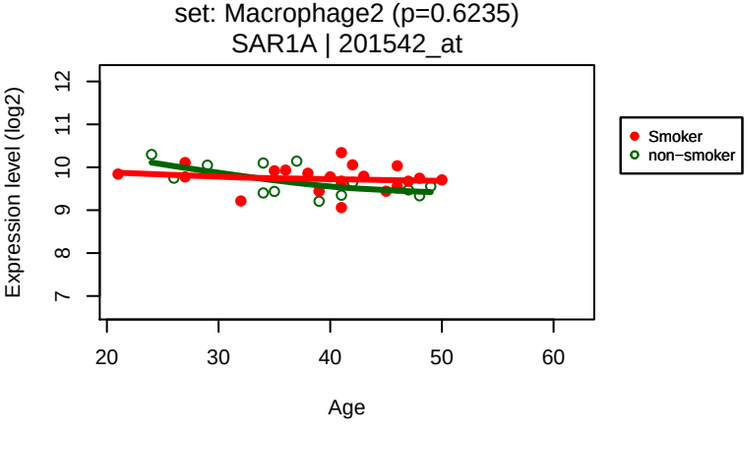

set: SmallAirway2 (p=0.3336)
SAR1A | 201542_at
Expression level (log2)
Age
Smoker
non−smoker

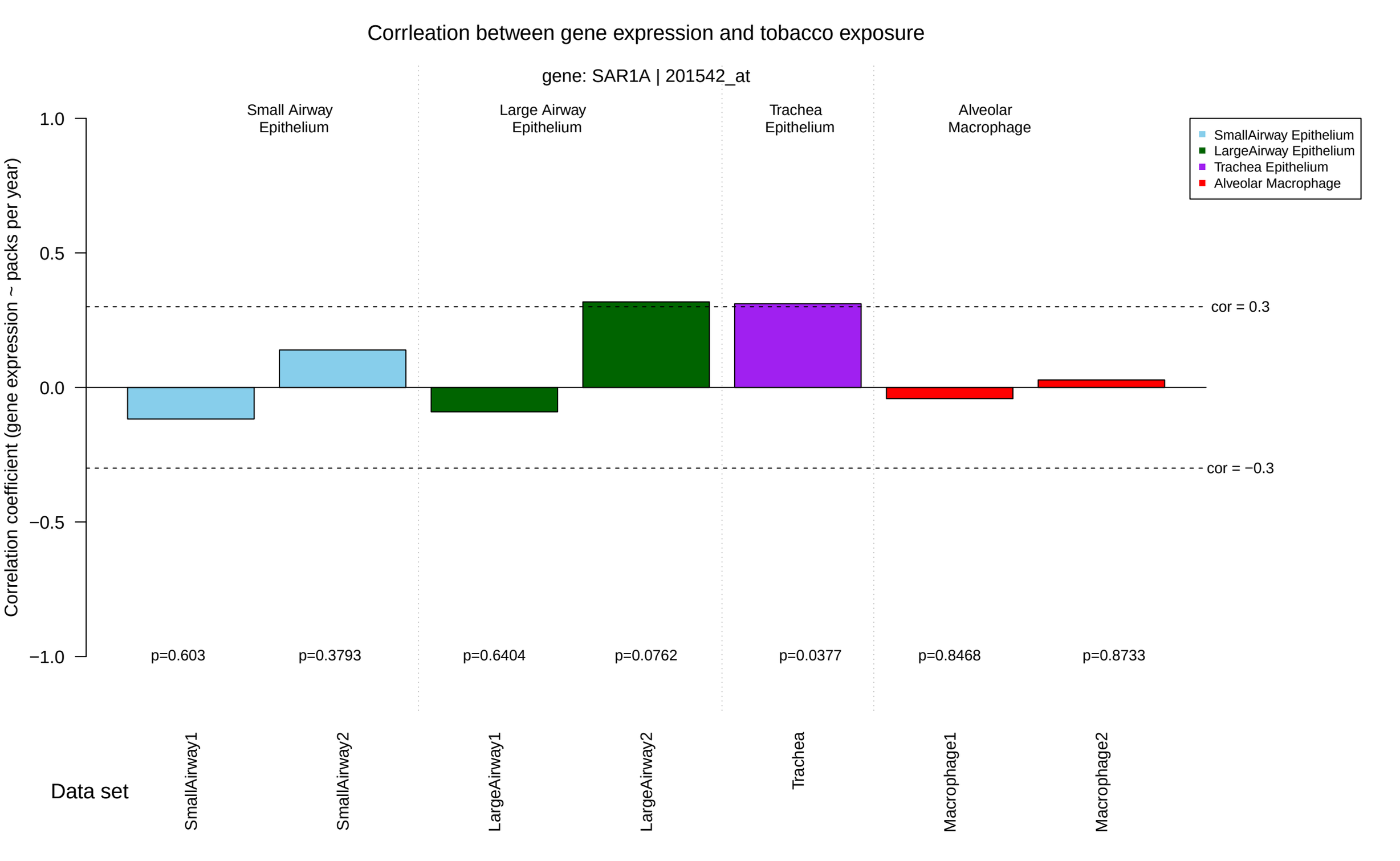

set: LargeAirway1 (p=0.3)
SAR1A | 201542_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.289)
SAR1A | 201542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0935)
SAR1A | 201542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7098)
SAR1A | 201542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6235)
SAR1A | 201542_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAR1A | 201542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.603
p=0.3793
p=0.6404
p=0.0762
p=0.0377
p=0.8468
p=0.8733
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2